Smart Mirror Enabled IoT Home Automation

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***Abstract*—This paper looks at home automation systems that use Internet of Things technology and have a smart mirror interface. It looks at how these systemsre set up how they talk to each other using things like MQTT how to put them in place what the security problems are and how to use less energy. The smart mirror is, like a screen that you can use to watch what is going on in your home and control your devices in real time.This paper also finds areas that need research and suggests a new way of doing things that combines local and cloud processing to make smart homes work better and be more secure..**

***Index Terms*—Internet of Things (IoT), Smart Home, Home Automation, Smart Mirror, System Architecture, MQTT, Security, Energy Optimization.**

I. Introduction

Smart home automation is about using Internet of Things technologies like sensors and communication networks to monitor and control things at home. This makes life easier and more comfortable. A lot of people are moving to cities. They want to save energy and be safe. They also want their homes to be comfortable. That is why people are doing a lot of research on homes. In fact the smart home market is going to be really big. It will be over 300 billion dollors by 2026. This means a lot of people are using home technologies.

Old home automation systems used wires to connect things. These systems were not very flexible. You could not control them from far away.. Now we have new systems that use the Internet of Things. These systems can monitor things in time and you can control them from your phone or computer. They can even predict what you want to do. However using your phone to control everything can be annoying. That is why a smart mirror can be an idea. A smart mirror is like a screen that shows you information like the temperature and weather. You can also use it to control your household devices. It is, like a control panel that's easy to use and looks nice. Smart mirrors can make it easier for people to use their homes.

* They can show you real-time information * They can control household devices * They are easy to use and look

Smart home automation and smart mirrors can make life easier and more comfortable. The smart home market is growing fast and more people are using home technologies. Smart mirrors are a way to control your smart home. They are easy to use and provide a lot of information.

II. Problem Definition

Despite rapid advancements in Internet of Things (IoT) technologies and their application in smart home environments, several challenges continue to hinder the widespread adoption and optimal performance of these systems. Many existing solutions focus primarily on remote device control rather than intelligent automation, resulting in limitations related to usability, efficiency, and security.

- **Lack of Centralized Interactive Interfaces:** Most existing smart home systems rely heavily on smartphone applications for device monitoring and control. This interaction method requires users to unlock their phones and navigate through multiple application menus, which reduces convenience in daily use. As a result, the absence of a dedicated, always-available interface such as a smart mirror or centralized dashboard limits seamless user interaction within the home environment.
- **Inefficient Energy Management:** Many current smart home systems function primarily as remote switching platforms rather than intelligent automation systems. They lack adaptive decision-making capabilities that can optimize energy consumption based on environmental conditions, user behavior, or occupancy patterns. Consequently, these systems often fail to fully utilize the potential of automation for energy efficiency.
- **Security Vulnerabilities:** Security vulnerabilities also remain a critical concern in IoT-based home automation networks. Such systems are susceptible to several cybersecurity threats, including Man-in-the-Middle (MITM) attacks, Distributed Denial-of-Service (DDoS) attacks, and unauthorized device access. These risks often arise from weak authentication mechanisms and the absence of encryption in lightweight communication protocols such as MQTT..

III. Design Goals

The proposed system architecture aims to address the aforementioned problems with the following specific design goals:

1) **Centralized Dashboard:**One of the main goals of the proposed system is to develop a centralized dashboard using a smart mirror interface. The smart mirror acts as an always-available interactive display that aggregates environmental data and device status while also enabling users to control household appliances in a seamless and non-intrusive manner.
2) **Scalable Architecture:**Another important objective is to design a scalable architecture capable of supporting multiple distributed IoT nodes. In this architecture, several ESP32 microcontrollers can be deployed in different areas of the house, such as the living room, kitchen, and bedroom. This multi-node topology allows the system to expand easily without causing network congestion or reducing system performance.
3) **Hybrid Control:**The system also aims to implement a hybrid control mechanism that combines both local network communication and cloud-based services. Local communication enables low-latency device control within the home network, while cloud connectivity allows remote monitoring and control through the internet. The MQTT protocol is utilized to ensure lightweight and efficient communication between devices.
4) **Enhanced Security:** . To protect the IoT network from unauthorized access and cyber threats, the architecture incorporates secure communication techniques such as TLS/SSL encryption along with strong authentication mechanisms. These measures help ensure reliable and secure operation of the smart home automation system.

IV. Literature Review

The evolution of IoT-based smart home systems has been widely explored in recent years. Madakam et al. [1] provided a foundational taxonomy of IoT architectures, identifying perception, network, and application layers as core structural components. However, their work focused primarily on conceptual classification rather than practical automation deployment.

Alheraish [2] implemented one of the earlier home automation frameworks using embedded systems, demonstrating feasibility but lacking scalability and cloud integration. Later, Yassein et al. [3] introduced MQTT-based communication for smart homes, highlighting its lightweight publish–subscribe mechanism and reduced network overhead compared to HTTP. Their results showed improved efficiency in constrained environments.

Security has been extensively studied by Ammar et al. [9] and Neshenko et al. [10], who emphasized the vulnerability of IoT ecosystems to attacks such as DDoS, spoofing, and unauthorized broker access. These works strongly recommend encrypted communication and secure authentication mechanisms.

Islam et al. [14] proposed a cloud-integrated smart home framework, improving remote accessibility but introducing latency due to dependence on cloud processing. Meanwhile, research on smart mirrors primarily focuses on user interface enhancement rather than full automation control integration.

Despite these contributions, most existing systems suffer from one or more of the following limitations:

- Lack of centralized interactive dashboard
- Absence of hybrid local-cloud control
- Limited integration of AI-based voice interaction
- Weak security reinforcement in MQTT deployments.

TABLE I: Comparative Analysis of Existing Smart Home Automation Systems

Ref	Author	Year	Focus	Protocol	Strength	Limitation
[1]	Madakam et al.	2015	IoT Architecture	HTTP	Well-defined IoT taxonomy	No practical implementation
[3]	Yassein et al.	2017	Smart Home System	MQTT	Lightweight communication	Limited security mechanisms
[9]	Ammar et al.	2015	IoT Security	Multiple	Security analysis	No automation architecture
[14]	Islam et al.	2021	Cloud Automation	MQTT	Remote monitoring	Cloud latency
–	Proposed System	2026	Smart Mirror Automation	MQTT + Hybrid	Central dashboard	Setup complexity

A. Research Gap Analysis

Although numerous IoT-based home automation systems have been proposed, many existing solutions still lack a unified platform that integrates device control, environmental monitoring, and information visualization within a single interactive interface. Most current systems primarily rely on mobile applications for device management, which limits usability and continuous accessibility within the home environment.

Furthermore, several smart home architectures depend heavily on cloud-based infrastructures. While cloud integration enables remote monitoring and data storage, it may introduce communication latency and reduce system reliability during network disruptions. In addition, security mechanisms implemented in many MQTT-based IoT systems remain insufficient, exposing devices to potential threats such as unauthorized access and data interception.

Another limitation in existing research is the lack of integration between intelligent user interfaces and home automation control systems. In particular, the use of smart mirrors as centralized dashboards for real-time monitoring and device interaction has not been extensively explored.

Therefore, there is a clear need for a scalable and secure smart home automation architecture that integrates the following key features:

- Smart mirror–based visualization and centralized control
- AI-enabled voice interaction for user convenience
- Secure MQTT communication using TLS encryption
- Hybrid local–cloud processing to reduce latency
- Intelligent energy optimization using environmental sensing

B. Performance and Scalability Analysis

1) Performance Considerations: The proposed architecture improves system performance by incorporating efficient com-

munication mechanisms and energy-aware automation strategies. The use of a local MQTT broker enables faster message delivery and reduces device switching latency, which is expected to remain below 200 ms. The publish–subscribe communication model further minimizes bandwidth consumption and ensures efficient message distribution among IoT nodes. In addition, the ESP32 microcontroller supports deep sleep modes that significantly reduce power consumption in idle states. Motion-based automation using PIR sensors also prevents unnecessary appliance operation, contributing to improved energy efficiency.

2) Scalability: The proposed architecture is designed to support scalable deployment across multiple rooms and devices. Each ESP32 node publishes and subscribes to unique MQTT topics, allowing independent communication with the central broker. The hierarchical topic structure of MQTT enables efficient message management, even when the number of connected devices increases. As a result, additional IoT nodes can be integrated into the system without modifying the core system architecture, allowing seamless expansion of the smart home network.

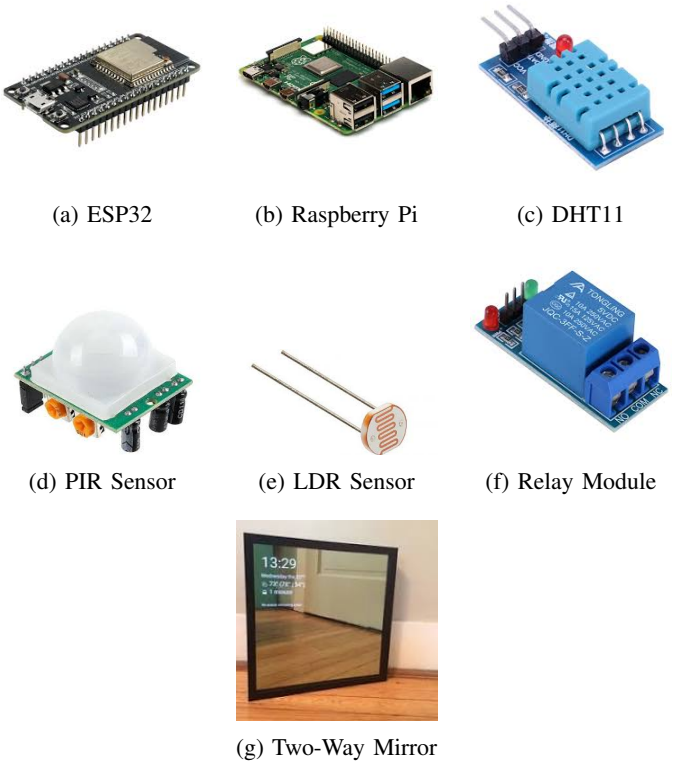
V. System Architecture

The proposed smart home automation system follows a layered Internet of Things (IoT) architecture consisting of three primary layers: the Perception Layer, Network Layer, and Application Layer. This layered approach enables modular system design, efficient communication between devices, and simplified scalability for future expansion. The architecture integrates distributed IoT sensor nodes, a centralized processing unit, and a smart mirror interface to enable real-time monitoring and control of home appliances.

A. Hardware Components

The selection of hardware components is critical for balancing performance, cost, and power efficiency in an IoT ecosystem.

- **ESP32 Microcontrollers:**The ESP32 (ESP-WROOM-32 module) serves as the core processing unit for distributed sensor nodes. It was selected due to its dual-core 240 MHz processor, integrated Wi-Fi and Bluetooth/BLE connectivity, and multiple GPIO interfaces. Compared to alternatives such as Arduino Uno or ESP8266, the ESP32 provides greater computational capability and improved connectivity. Additionally, its deep sleep mode allows for energy-efficient operation, making it suitable for IoT-based sensor deployments.
- **Raspberry Pi 4 Model B:**The Raspberry Pi 4 acts as the central processing unit and smart mirror controller. It features a quad-core Cortex-A72 processor and supports dual 4K video outputs. The device hosts the MagicMirror[2] framework, MQTT broker services, and Node.js-based backend modules. This allows the smart mirror to process incoming sensor data and display real-time information through an interactive interface.
- **DHT11 Sensor:**The DHT11 sensor is used to measure environmental parameters such as temperature and humidity within the smart home environment. It provides a temperature range of 0–50°C and humidity levels between 20–90
- **PIR Motion Sensor (HC-SR501):** The PIR sensor detects human presence by sensing changes in infrared radiation emitted by the human body. This sensor is primarily used for occupancy-based automation tasks, such as automatically turning on lights when motion is detected in a room.
- **LDR (Light Dependent Resistor):** The LDR sensor measures ambient light intensity within the environment. This data enables adaptive control of lighting systems and allows the smart mirror display brightness to adjust dynamically according to surrounding light conditions.
- **Relay Modules:** Relay modules act as switching interfaces between low-voltage microcontroller signals and high-voltage household appliances. These relays allow the ESP32 nodes to safely control devices such as lights, fans, and other electrical appliances.
- **Two-Way Mirror:**The smart mirror uses a two-way reflective glass that functions as both a mirror and a display surface. When combined with a monitor placed behind the glass, the mirror creates a transparent display effect where digital information appears to float on the mirror surface.



(a) ESP32 (b) Raspberry Pi (c) DHT11

(d) PIR Sensor (e) LDR Sensor (f) Relay Module

(g) Two-Way Mirror

Fig. 1: Hardware components used in the proposed smart home automation system

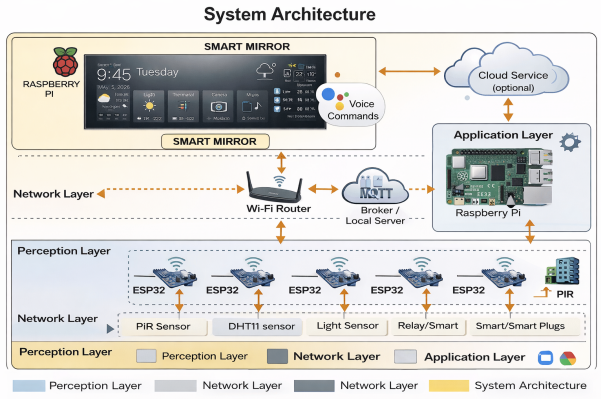


Fig. 2: Architecture

B. Network Topology

The system follows a star network topology in which multiple ESP32 sensor nodes communicate with a central server hosting the MQTT broker. Each ESP32 node publishes sensor readings to designated MQTT topics and subscribes to control topics for device actuation. The Raspberry Pi–based smart mirror subscribes to sensor data topics and publishes control commands when users interact with the interface. This communication model enables efficient message exchange using the lightweight MQTT publish–subscribe protocol.

C. System Architecture Diagram

- **Perception Layer:** Contains ESP32 nodes, sensors (DHT11, PIR, LDR), and relay modules responsible for sensing environmental data and controlling appliances.
- **Network Layer:** Includes Wi-Fi communication infrastructure and the MQTT broker responsible for managing message exchange between IoT devices.
- **Application Layer:** Consists of the Raspberry Pi–based smart mirror interface that processes sensor data and provides a centralized dashboard for monitoring and device control.

D. Smart Mirror Internal Architecture

The internal architecture of the smart mirror integrates both hardware and software components to enable interactive visualization and automation control. As illustrated in Figure **??**, the hardware stack includes a display monitor mounted behind a two-way mirror panel and powered by a Raspberry Pi computing unit.

On the software side, the Raspberry Pi runs the MagicMirror[2] framework along with Node.js backend modules that handle MQTT communication and data processing. Custom modules subscribe to sensor topics and dynamically update the mirror interface with real-time information such as environmental conditions, appliance status, and system notifications. This integration enables the smart mirror to function as a centralized control dashboard for the entire smart home ecosystem.

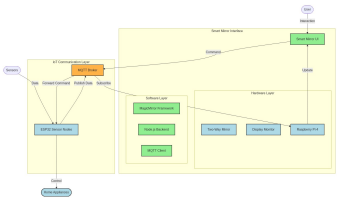


Fig. 3: mirror architecture

VI. Workflow Execution

The proposed Smart Mirror–based Home Automation system operates using a continuous cyclic workflow where sensing, communication, visualization, and control occur simultaneously. The system integrates IoT devices, cloud messaging, and a smart interface to ensure real-time automation and user interaction.

A. System Initialization

When the system powers ON, all hardware components begin their startup procedures.

- The ESP32 microcontroller initializes GPIO pins, sensors, and relay modules.
- It connects to the configured WiFi network using stored credentials.
- After successful internet connectivity, the ESP32 establishes communication with the MQTT Broker.
- A handshake process verifies device identity using client ID and authentication credentials.
- Simultaneously, the Raspberry Pi boots the Smart Mirror software environment and launches the MagicMirror[2] dashboard.

This step ensures that all devices are connected within the same IoT communication network before data exchange begins.

B. Data Acquisition (Sensing Phase)

Once initialized, sensors start monitoring environmental parameters continuously. Typical sensors include: Temperature & Humidity Sensor, Gas/Air Quality Sensor, Light sensor (LDR), and Motion sensor (PIR). The ESP32 reads sensor values at predefined intervals. Analog signals are converted into digital values using internal ADC processing. The collected data represents real-time environmental conditions inside the home.

C. Data Transmission via MQTT

After acquiring sensor readings:

- ESP32 formats the data into structured JSON payloads.
- These payloads are published to specific MQTT topics like `home/sensors/environment`.

The MQTT publish–subscribe model ensures lightweight communication, low bandwidth usage, and fast message delivery. The broker acts as an intermediary, distributing data to subscribed devices.

D. Smart Mirror Display & Visualization

The Smart Mirror (running on Raspberry Pi) subscribes to relevant MQTT topics. When new data arrives:

- Node.js backend processes incoming messages.
- MagicMirror^2 modules update the dashboard instantly.
- Users see live information such as temperature, weather, device status, notifications, time, and schedule reminders.

This creates a centralized visual control panel embedded inside the mirror surface.

E. User Control & Device Actuation

The system supports bidirectional communication. When a user interacts with the mirror using touch input, voice command, or UI buttons, the Smart Mirror publishes control commands back to MQTT topics such as `home/control/light1`. The ESP32 subscribes to these control topics and:

- Receives the command
- Processes instruction logic
- Activates relay modules
- Turns appliances ON/OFF accordingly

This completes one automation cycle.

F. Continuous Cyclic Operation

The workflow repeats continuously: Sensors $\rightarrow$ ESP32 $\rightarrow$ MQTT Broker $\rightarrow$ Smart Mirror. Commands flow back from User Interaction to the mirror, then to the broker, and finally to the ESP32. This cyclic architecture enables real-time monitoring, instant response, remote automation, and scalable device addition.

G. Example Execution Scenario

1) Temperature increases in the room.
2) Sensor detects change and sends data via ESP32.
3) Smart Mirror updates temperature display instantly.
4) User says: "Turn ON fan."
5) Mirror sends MQTT command.
6) ESP32 activates relay.
7) Fan turns ON automatically.

VII. Implementation Guidelines

The implementation of the Smart Mirror–based Home Automation system involves integrating multiple software platforms and hardware components to achieve seamless communication, real-time monitoring, and device control. The development process follows a structured setup.

A. Software Environment Setup

The system relies on several software tools, each responsible for a specific function within the architecture.

- **Arduino IDE:** Used for programming and uploading firmware to the ESP32 microcontroller. It enables sensor data acquisition, WiFi connectivity, and MQTT communication logic.
- **Raspberry Pi OS:** Acts as the operating system for the Raspberry Pi, providing a Linux-based environment required to run automation services and Smart Mirror applications.
- **MagicMirror^2 Framework:** Provides the graphical interface displayed on the mirror. It allows modular dashboard development including widgets for time, weather, notifications, and device status.
- **Mosquitto MQTT Broker:** Handles message exchange between ESP32 devices and the Smart Mirror using a publish–subscribe communication model.
- **Node.js Runtime:** Executes backend scripts and custom MagicMirror modules for processing MQTT data and updating the interface dynamically.

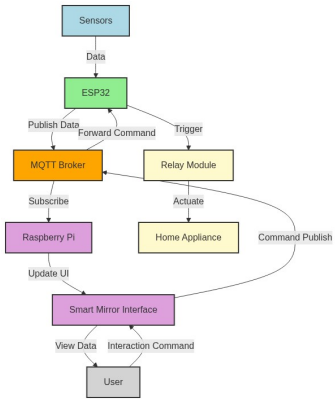


Fig. 4: Workflow execution of the proposed smart mirror-based home automation system

B. Network Configuration

Proper network setup ensures stable communication among all devices. Implementation steps include:

- Assigning static or reserved IP addresses to Raspberry Pi (Smart Mirror), MQTT Broker, and ESP32 devices.
- Connecting all devices to the same WiFi network.
- Testing connectivity using ping commands.
- Configuring MQTT broker address inside ESP32 firmware.

C. Sensor Integration

Sensors are connected to ESP32 GPIO pins based on required input types. Implementation process:

1) Connect sensors using appropriate voltage levels.
2) Install necessary sensor libraries in Arduino IDE.
3) Write firmware to read sensor values periodically.
4) Calibrate sensors for accurate readings.
5) Convert raw values into meaningful environmental data.

D. MQTT Communication Setup

Communication between devices is established using MQTT protocol. Steps include:

- Installing Mosquitto broker on Raspberry Pi.
- Creating MQTT topics for sensor data publishing and device control commands.
- Programming ESP32 to publish and subscribe to topics.

- Configuring Smart Mirror modules to subscribe to sensor topics.
- Testing message flow using MQTT client tools.

E. Smart Mirror Module Development

Custom modules are developed inside the MagicMirror[2] framework. Implementation tasks:

- Designing dashboard layout using HTML, CSS, and JavaScript.
- Creating Node.js backend scripts.
- Subscribing to MQTT topics for live data updates.
- Implementing UI elements for appliance control.
- Adding voice command integration (optional AI assistant).

F. Security Validation

Security is essential in IoT systems to prevent unauthorized access. Validation measures include:

- Enabling MQTT authentication (username/password).
- Using encrypted communication where possible.
- Restricting device access through firewall rules.
- Securing WiFi network credentials.
- Testing unauthorized login attempts.

G. System Testing and Deployment

After integration, the system undergoes testing:

- Sensor data accuracy testing
- Command execution delay measurement
- Network reliability checks
- End-to-end workflow verification

Once validated, the Smart Mirror is mounted and deployed for real-world operation.

VIII. Security Considerations

Security is a critical requirement in IoT-based smart home systems to protect devices, communication channels, and user data from unauthorized access and cyber threats. The proposed system incorporates several security practices to ensure reliable and secure operation:

- **Encrypted Communication:** ISecure communication is implemented using TLS/SSL encryption for MQTT data transmission. This prevents attackers from intercepting or eavesdropping on sensitive data exchanged between IoT devices and the smart mirror interface.
- **Authentication:**Strong authentication mechanisms are used to control access to the MQTT broker. Each IoT device is assigned a unique client ID along with username and password credentials to prevent unauthorized device connections.
- **Network Protection:** Firewall rules and port filtering are applied to restrict access to critical services such as the MQTT broker. Only authorized devices within the local network are permitted to communicate with the system.
- **Secure Firmware Updates:** Over-The-Air (OTA) firmware updates are used to regularly patch vulnerabilities and ensure that ESP32 nodes run secure and updated firmware versions.
- **Wi-Fi Security:** The wireless network used by the IoT devices is secured using WPA2/WPA3 encryption to prevent unauthorized network access and protect data transmission.

These security measures collectively reduce the risk of common IoT attacks such as unauthorized access, message interception, and service disruption.

TABLE II: Security Threat & Mitigation Analysis

Threat	**Impact**	**Mitigation Strategy**
MITM Attack	Data interception	TLS Encryption
Unauthorized Access	Device control breach	Strong authentication
DDoS	Broker overload	Firewall + Rate limiting
Credential Theft	System compromise	Token-based authentication

IX. Energy Optimization

Energy efficiency is an important consideration in smart home automation systems, particularly for reducing electricity consumption and improving sustainable operation. The proposed system incorporates several automation strategies to optimize energy usage within the home environment.

Motion-based automation using PIR sensors enables automatic shutdown of lighting and appliances when no human presence is detected for a predefined duration. Additionally, temperature-based fan control can be implemented using PWM (Pulse Width Modulation) to dynamically adjust fan speed according to environmental conditions, thereby reducing unnecessary energy consumption.

Scheduling mechanisms can also be applied to manage high-power appliances during off-peak hours. Furthermore, integrating current sensors such as the ACS712 enables real-time monitoring of power consumption. The collected energy usage data can be visualized on the smart mirror interface, allowing users to observe their electricity consumption patterns and adopt more energy-efficient behaviors.

X. Future Research Directions

Although the proposed smart mirror–based home automation system provides an effective framework for intelligent home management, several research opportunities remain for further enhancement.

Future developments may include the integration of artificial intelligence techniques for predictive automation, enabling the system to learn user behavior patterns and automatically optimize device operation. Blockchain-based security mechanisms may also be explored to

provide decentralized trust and improve the protection of IoT communication networks.

Edge computing can be incorporated to enable local data processing and reduce dependency on cloud infrastructure, thereby improving system responsiveness and reducing latency. In addition, biometric authentication methods such as facial recognition can be implemented on the smart mirror to provide personalized user profiles and secure system access.

Another promising direction involves applying machine learning algorithms to analyze historical energy consumption data and generate accurate energy usage forecasts. Such predictive models could further enhance energy optimization and support intelligent decision-making in smart home environments.

XI. Conclusion

This paper presented a comprehensive review and system architecture for an IoT-based smart home automation framework integrated with an interactive smart mirror interface. The proposed approach combines IoT sensor networks, MQTT-based communication, and a centralized smart mirror dashboard to enable real-time monitoring and intelligent control of household appliances.

The architecture leverages ESP32 microcontrollers for distributed sensing and device control, while a Raspberry Pi–based smart mirror acts as the central user interface. The use of the MQTT publish–subscribe protocol enables lightweight and low-latency communication among IoT nodes, ensuring reliable operation even in resource-constrained environments. In addition, the implementation guidelines demonstrate how the integration of hardware components, communication protocols, and software frameworks can support a scalable smart home ecosystem.

Security considerations such as encrypted communication, authentication mechanisms, and secure network configurations are highlighted to address common vulnerabilities in IoT deployments. Furthermore, energy optimization strategies—including motion-based automation, intelligent scheduling, and real-time energy monitoring—illustrate the potential of the proposed system to promote efficient and sustainable energy usage in residential environments.

Although the system introduces an effective framework for interactive smart home automation, certain challenges remain, including initial deployment complexity and the need for continuous security maintenance. Future enhancements may involve integrating edge computing capabilities, AI-based predictive automation, and biometric authentication mechanisms to enable personalized and intelligent home environments.

Overall, the integration of IoT technologies with intuitive interfaces such as smart mirrors represents a promising direction for next-generation smart home systems, offering improved usability, efficiency, and intelligent automation for modern living environments.

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